



Assessing the soil moisture effects of planted vegetation on slope stability in shallow landslide-prone areas

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Received: 16 December 2020 / Accepted: 18 April 2021 / Published online: 23 April 2021
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Abstract

Purpose Soil moisture plays an important factor to impact runoff and the slope stability, but how vegetation affects the soil moisture variability of slope stability is poorly understood. In order to compare the stability of different shrub slopes under the effect of soil moisture, four types of shrubs (*Periploca sepium*, *Lespedeza bicolor*, *Ziziphus jujuba*, and *Punica granatum*) are selected for ecological restoration in shallow landslide-prone areas, and the effects of rhizosphere soil moisture and plant roots on the slope stability are studied.

Materials and methods Trace infiltration is applied in order to study plant roots and soil moisture impact. The shear strength of the roots slope soil under different moisture content is determined by laboratory shear strength test. The slope safety factor of different shrubs is obtained by FLAC 3D (Fast Lagrangian Analysis of Continua 3D) simulation.

Results and discussion Among the four shrubs, thick roots show the greatest improvement to soil porosity, and fine and medium roots could provide more preferential flow channels for water infiltration. The soil moisture content significantly increases after planting *Lespedeza bicolor*. At 10% soil moisture content, the *P. sepium* slope safety factor is highest in four shrubs (1.39). When the soil moisture content reaches 30%, the safety factor value of the *L. bicolor* slope reaches the maximum, and the safety factor value of *L. bicolor* decreased by 25.6%.

Conclusions The results show that even during the wettest condition of the soil (saturated soil moisture content 30%), the mechanical reinforcement from the *L. bicolor* root maintains some degree of stability. Shrubs have a good effect on enhancing the stability of slopes, especially in areas prone to shallow landslides.

Keywords Shallow landslides · Soil moisture · Root system · Slope stability

1 Introduction

Almost a fifth of global land surface is classified as highly susceptible to shallow, rainfall-induced landslides (Keefer and Larsen 2007). The presence of herbaceous plants can reduce soil erosion on slopes, because plant roots affect soil properties such as soil moisture, soil porosity, soil shear strength, and preferential flow characteristics. Vegetation, soil, and their composites are the basis for the maintenance and development of slope ecosystems (Gonzalez-Ollauri and Mickovski 2016). Soil moisture is an significant basis for plant growth, and also an important factor influencing surface runoff and even shallow landslides (Lipiec et al. 2006). And vertical soil moisture profiles at selected locations represent the main features of the spatial variation of the catchment's

Responsible editor: Saskia D. Keesstra

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wetness (Chiffard et al. 2018). Similarly, Bajwa et al. (2017) find that different plant species respond differently to soil moisture stress, and the higher root to shoot length ratios observed under low soil moisture holding capacity may be another key feature of tolerance to soil moisture deficiency. Planted vegetation has both mechanical and hydrological impacts on slope stability, some of which can improve and some of which could decrease the slope stability. Thus, the need for sound scientific evidence on how soil moisture affects slope stability is urgently.

Hydrological conditions on vegetation slopes are affected by soil structure and plant root morphology (Sidle et al. 2006; Pierret and Moran 2011; Bengough 2012). When plant roots grow in the soil, stable passages are generated mainly by compressing soil particles and merging roots and liquids as they pass through the soil (Bengough et al. 1997). Due to gravity, soil moisture moves down the soil profile through these channels. Because of the uneven distribution of soil particles and the extension of the root system, the distribution of soil moisture in the profile is not uniform (Bodner et al. 2014). Plant roots have both positive and negative effects on water distribution on slopes (Ghestem et al. 2011). On the one hand, they can promote water infiltration and improve slope stability (Noguchi et al. 1999; Noguchi et al. 2001). Preferential flow is defined as the physical movement of water and solutes along a certain path, bypassing part of the pore matrix (Hendrickx and Flury 2001). Preferential flow is a common form of soil water movement. Dye tracer experiments can show preferential flow paths in soil (Flury and Flühler 1995). Subsequent analysis of the original stained image by PS and MATLAB yields specific staining parameters such as the proportion of stained area, depth and width of staining. On the other hand, water accumulates at points where the root system gathers, leading to an increase in water pressure and sometimes even reaching the critical value of the slope, causing slope instability (Nespoulous et al. 2018). It remains to be studied how different vegetation types and plant root distribution affect soil pores, water distribution and preferential flow paths. Therefore, the change of soil moisture is of great importance in the study of the slope stability in shallow landslide-prone areas.

Vegetation roots can play an effective role in suppressing landslides through reinforcement or anchoring (Stokes et al. 2007). The additional cohesive force that plant roots can produce on the soil enhances the shear strength of the soil and strengthens the soil. When the moisture content of the rhizosphere soil changes, soil cohesion will be affected, and the soil shear strength will change accordingly. During the implementation of plant slope protection measures, the spatial distribution of plant roots has an effect on the effect of slope stabilization (Ji et al. 2012). Therefore, it is important to explore the heterogeneity of the spatial distribution of roots under different restoration shrub species.

Soil shear strength is an important factor in evaluating slope stability (Lian et al. 2019). When the shear strength is small, it is not conducive to the stability of the soil. Conversely, when the soil stabilization effect is poor, slope instability and shallow landslides are likely to occur. There are many factors that affect the shear strength of soil. The main influencing factors are soil cohesion, internal friction angle, and so on (Stokes et al. 2009; Li et al. 2019a). In the same research site, the soil moisture content and the type of root system mainly affect the shear strength of the soil. The change of soil moisture content will affect the cohesion and internal friction angle of the soil, and then change the shear strength of the soil. A series of studies have been carried out on the effects of plant roots and soil moisture on soil shear strength, including field tests, laboratory tests and model analysis (Wu and Watson 1998; Comino and Druetta 2010; Thomas and Pollen 2010; Fan and Tsai 2016). In situ shear tests are performed on soil containing roots, and the results show that the shear strength of rhizosphere soil is higher than that of none roots soil, the roots of plants could reinforce the soil (Wu and Watson 1998). Extensive work has been performed in the laboratory on remodeled soils with different moisture contents and different root systems through direct shear tests to obtain optimal soil moisture contents and plant root systems that enhance soil shear strength (Ghestem et al. 2014). In order to assess the impact between soil moisture dynamics and plant ecology, Nimmo et al. (2009) conduct infiltration and water redistribution experiments, and the study concludes that water infiltration is beneficial to deep-rooted plants. Vegetation can change soil properties on a time scale, affect root water supply, reduce evaporation and drainage losses (Nimmo et al. 2009; Sandvig and Phillips 2006). Changes in land use patterns can also significantly affect soil hydraulic properties (Wahren et al. 2009; Zimmermann et al. 2006).

The numerical simulation method is a widely used method for slope stability analysis (Kokutse et al. 2016; Temgoua et al. 2017). The numerical simulation method focuses on the analysis of soil stress-strain or failure mechanism. The strength reduction method based on numerical simulation technology can directly obtain the slope safety factor (FoS) without assuming the shape and position of the sliding surface in advance, and can well reveal the slope slip law. This method has been widely used in slope stability analysis in recent years (Kokutse et al. 2016; Temgoua et al. 2017). At present, new models and tools are mainly used and developed with less data, in order to evaluate and implement suitable management programs, and to better explain the model of modeling phenomena (Le et al. 2012; Rhodes 2014). In the 1980s, the American ITASCA company launched the Fast Lagrangian Analysis of Continua 3D (FLAC3D) software. The software can well build slope models and input related parameters, and the potential sliding surface can be calculated and estimated by implementing strength reduction, and the FoS can be

obtained. This method can analyze the change and the whole process of soil stress and strain. The strength reduction method based on numerical simulation can directly obtain the safety factor of the slope without assuming the shape and position of the sliding surface in advance, which can well reveal the processes of movement and sliding on the slope. In this study, after obtaining shear strength data through laboratory direct shear tests, numerical simulations are performed on the corresponding slopes to obtain corresponding factor of slope safety. The effect of slope stability under different soil moisture and vegetation conditions is displayed, and the main factors affecting the safety factor are analyzed to provide a basis for the design of the later slope management scheme.

The aim of this study is to provide a basis for the selection of the dominant shrub species for the implementation of slope protection in shallow landslide-prone areas. To this purpose, four typical shrubs are selected in the shallow landslide-prone area of one of the four major geological hazard-prone areas in China, and the plant species and plots are investigated in the field. In the laboratory, the direct shear test is carried out. Finally, the FLAC3D software is used to simulate the slope safety factor of different shrubs with different water contents using the strength reduction method. The effects of plant roots and soil moisture on slope stability are comprehensively analyzed. This study will help to provide a reference for the selection of suitable vegetation types for ecological control of shallow landslides in the future.

2 Study sites

In order to reveal the influence of soil moisture on the stability of slopes in shallow landslide-prone areas, this study selects the Longnan section of the Bailong River Basin in southern Gansu province, China as the study area. This area is the intersection of the three major regions: the Western Qinling Mountain, the Qinghai-Tibet Plateau, and the Sichuan Basin (Meng et al. 2013). It belongs to the erosion and denudation tectonics of high mountains, and the type of deep cut canyons in high mountains (Wang et al. 2016). It is also an important ecological barrier in the upper reaches of the Yangtze River in China and an important source of water protection in Gansu province (Wang et al. 2019). In recent years, due to vegetation degradation and frequent earthquakes, a large amount of loose material has been produced on the ground surface, and the risk of landslides has increased (Xiong et al. 2016). Wang et al. (2019) find 3,300 potential geological hazards in Longnan city, including 1,120 landslides, 1033 debris flows, 759 collapses, ground collapses, and ground cracks. During the reconstruction of the Wenchuan earthquake in 2008, a large number of vegetation restoration projects are carried out in the study area (Xiong et al. 2016; Wang et al. 2016).

The shrub vegetation hillside in the Duanheba Ecological Restoration Demonstration Area, which has been in the area for 10 years for ecological restoration, is selected as the research object (Fig. 1(a), (b)). The climate in the study area belongs to subtropical monsoon climate and temperate continental climate. The area has a lot of heavy rainfall from June to August, accounting for about 60% of the annual precipitation in the area. The average annual precipitation is about 700–900 mm. In addition, the precipitation shows an increasing trend with the increase of altitude. The annual average temperature is between 8.4 and 14.9 °C (Wang et al. 2019). The annual sunshine duration is about 1911 h, with sufficient light. The frost-free period is about 150–240 days.

The altitude is about 1500 m above sea level, the slope is about 30°. The main landform types in the study area include mountain landforms and valley landforms. The mountainous landforms are mainly mid-alpine landforms, which provide favorable topographic and landform conditions for the occurrence of landslides (Wang et al. 2016). Loess landforms are distributed in relatively flat and low-lying areas. Due to frequent tectonic activities in the study area, landslides, collapses, and other gravity erosion landforms are widely distributed. Such landslides are generally large in depth and scale (Zhao et al. 2020). The Alfisols is the main soil type in the study area, which is widely distributed in the mountain slopes, forests and shrubs of the study area (Xiong et al. 2016). For the potential landslides, soil erosion and other geological hazards in the study area, artificial restoration of afforestation area of about 2500 km² has been completed. The valley at low altitude and mountain slopes are mainly agricultural land and artificially restored ecological zones (Fig. 1(c)). The ecological zone is mainly composed of shrubs and herbaceous plants, which is convenient for studying the effect of different shrub types on soil and slope protection.

3 Materials and methods

3.1 Experimental design

Slopes with different dominant shrub species for vegetation restoration in this area are taken as the focus of the study. Four different restoration species are considered: *Periploca sepium*, *Lespedeza bicolor*, *Ziziphus jujuba*, and *Punica granatum*. Sampling is conducted in June 2018. Three sampling points are selected for each shrub, and bare land with similar site conditions as the sampling points is set as a control, and the sample area is 10 m × 10 m (Table 1). Soil samples are collected on the soil profile. Three repeated samples are taken at each sampling point, which are distributed in an S shape. On the vertical section of each shrub sampling point, soil samples are taken every 10 cm to a sampling depth of 50 cm. Mixing three soil samples at the same depth to obtain a typical soil

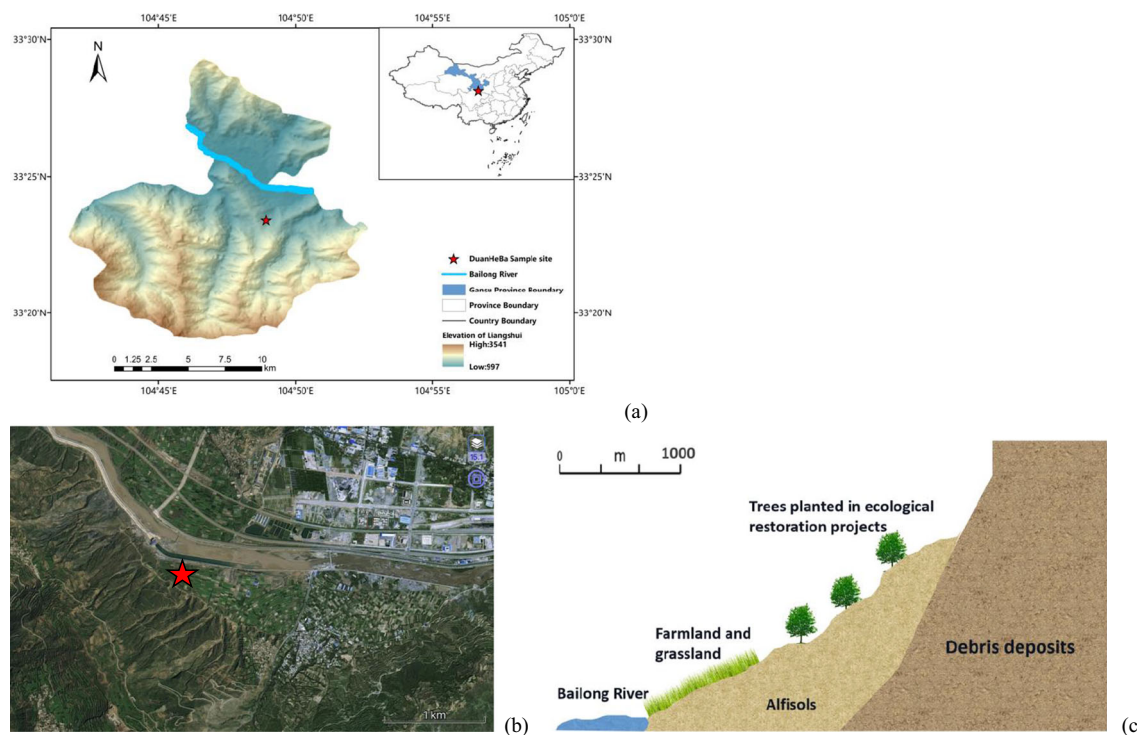


Fig. 1 (a) is location of the study sites within the area of debris flow in Bailong river basin, Longnan, southern of Gansu province, China. (b) is a topographic map generated using Google Earth. (c) is schematic cross-

sectional view of the research site of the Duanheba Ecological Restoration Demonstration Area

sample at this sampling site. The soil samples are sealed and stored in Ziplock bags. The soil porosity is calculated on site by ring knife measurement (Ring knife diameter: 50 mm; height: 50.46 mm, volume: 100 cm³). The soil moisture content is obtained by drying. Put the soil samples collected in the field into an aluminum box and weigh them. Dry at 105 ° for 10 h. After returning to room temperature, weigh again. Calculate the soil moisture content using the difference between dry and wet weight (Zhang et al. 2005).

The spatial distribution of the root system is analyzed (Schmidt et al. 2001; Sudmeyer et al. 2004). The root area ratio (RAR) was investigated using the vertical trench profile method in areas down-slope of the plants (Ji et al. 2012). After flattening, a 50 cm × 50 cm (10 cm × 10 cm interval) metal frame is placed to visually record the vertical distribution of the root system (Fig. 2). This method makes it easy to understand the root depth and root density distribution (Bouillet et al. 2002).

3.2 Dye tracer infiltration

Dye tracer experiments are carried out at the corresponding sampling points, and one plot is selected for each shrub vegetation type. A square with a side length of 60 cm is selected and a PVC hollow cylinder is placed. A bright blue solution (Flury and Flühler 1995) with a concentration of 5 g/L and a volume of 13 L is applied, and a thin film is used to cover the

surface of the cylinder to prevent water evaporation. After 24 h of infiltration, vertical and horizontal sections are excavated in the center of the cylinder. A digital camera is used to photograph the dyed soil profile for further data processing.

3.3 Shear strength test

In this study, the natural moisture content and saturated moisture content of the rhizosphere soil were used to determine the range of the remodeled soil moisture content (10–30% in intervals of 10%, 20%, and 30%). Remodeled rhizosphere soil samples are prepared at different moisture contents. First, the soil used to prepare the sample is dried to constant weight, ground through a 1-mm sieve, spread in a non-absorbent aluminum pan, and sprayed with the expected amount of water. The sample is then sealed with plastic wrap and moistened all day and night before use. The formula for adding water is as follows (Nanjing Hydraulic Research Institute 1999):

$$m_w = \frac{m}{1 + 0.01w_0} \times 0.01(w' - w_0)$$

where m_w is the added water required (g), m the mass of the soil sample (g), w_0 is the air-drying moisture content (%), and w' is the designed moisture content (%).

A strain-controlled direct shear apparatus (Nanjing Ningxi Soil Instrument Co., Ltd., Nanjing, Jiangsu, China) is used to determine the shear strength of the sample (Fig. 2(b)). Four

Table 1 Basic information of sampling site

Plant types	Site number	Latitude and longitude	Altitude (m, above sea level)	Angel of slope (°)	Plant height (cm)	Number of shrubs in the site
<i>Punica. Sepium</i> (<i>P. Sepium</i>)	1	N 34° 36' 50" E 105° 42' 52"	1380	24	121.8	11
	2	N 34° 22' 10" E 105° 38' 46"	1580	27	133.1	9
	3	N 34° 52' 09" E 104° 21' 19"	1189	20	149.6	10
<i>Lespedeza. Bicolor</i> (<i>L. Bicolor</i>)	1	N 33° 23' 40" E 104° 49' 08"	1172	22	153.3	9
	2	N 33° 32' 51" E 104° 38' 49"	1530	25	169.5	9
	3	N 33° 32' 1" E 104° 38' 20"	1258	20	158.3	11
<i>Ziziphus. Jujube</i> (<i>Z. Jujube</i>)	1	N 33° 32' 06" E 104° 37' 39"	1290	20	139.9	8
	2	N 33° 24' 22" E 104° 48' 43"	1100	18	142.8	9
	3	N 33° 32' 06" E 104° 38' 41"	1196	21	135.6	8
<i>Periploca. Granatum</i> (<i>P. Granatum</i>)	1	N 33° 12' 56" E 104° 25' 16"	1231	16	201.9	10
	2	N 33° 26' 37" E 103° 22' 52"	1428	20	194.6	9
	3	N 34° 13' 19" E 104° 20' 35"	1176	19	211.3	7
Bare land	1	N 33° 22' 54" E 104° 48' 49"	1368	18	—	—
	2	N 32° 23' 36" E 103° 20' 39"	1243	20	—	—
	3	N 32° 53' 12" E 103° 19' 01"	1103	23	—	—

samples from each group are tested under four vertical pressures of 100, 200, 300, and 400 kPa. Each treatment is repeated in triplicate.

3.4 Data analysis

To process the stained sections taken in the field, Adobe Photoshop CS6 (Adobe Systems Software Ireland Ltd, US)

is applied to transform the image into black and white then MATLAB is used to process the black and white image and output a binary image. By counting the number of stains in each 1 cm depth range, the change curve of staining area with depth was obtained (Wang et al. 2006).

The spatial distribution of root system under different recovery models was analyzed by the root area ratio (RAR). According to the specific conditions of the study area and

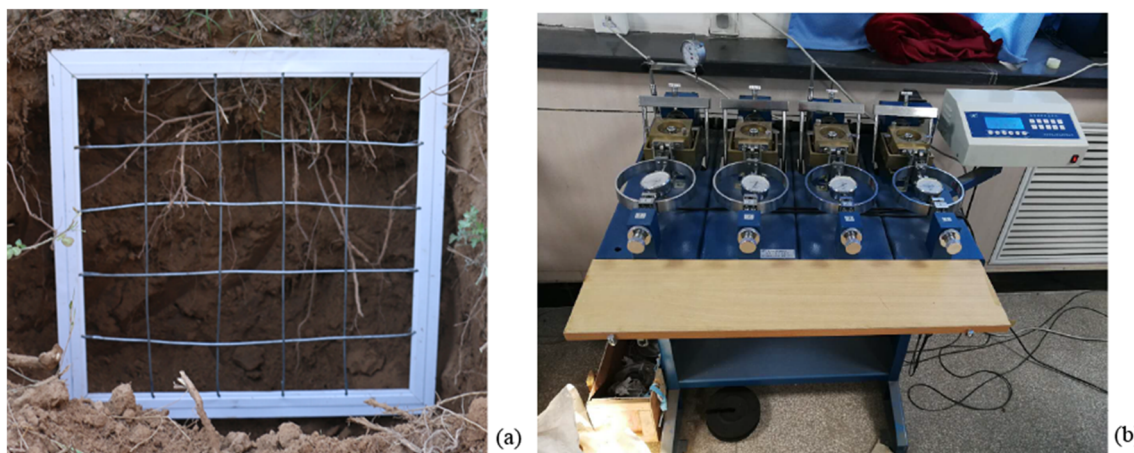


Fig. 2 (a) Recording the spatial distribution of plant roots by profile method. (b) A strain-controlled direct shear apparatus used to determine the shear strength of a sample (Nanjing Ningxi Soil Instrument Co., Ltd.)

the characteristics of the root system, the root system is divided into three diameter classes of 0–0.5, 0.5–2, and >2 mm (Drexhage and Gruber 1998; Sudmeyer et al. 2004; Genet et al. 2008; Ji et al. 2012). The RAR of the root system is the main parameter of the distribution pattern of the root diameter on the soil profile. The calculation formula is as follows:

$$RAR = \sum_{i=1}^n \frac{\pi D_i^2}{4A} = \sum_{i=1}^n \frac{\pi D_i^2}{4a \times b}$$

where D is the diameter of the root; A is the area of the shear plane; a is the horizontal length of the soil profile; and b is the vertical depth of the soil profile.

The Mohr-Coulomb strength theory follows the soil shear strength and vertical positive pressure (Ji et al. 2012):

$$\tau = C + \delta \tan \varphi$$

where τ is shear strength, C is effective cohesion, δ is vertical pressure, and φ is effective internal friction angle.

Using the C and φ values, the strength reduction method of the FLAC3D software is applied to continuously reduce the shear strength-related indexes of the sliding surface under study until the critical state of slope instability is reached. The reduction factor is the safety factor of the slope.

One-way analysis of variance is applied to evaluate the effect of roots on the dye infiltration parameters. The correlation coefficient is applied to indicate the correlation between the measured parameters. A significance level of $p < 0.05$ was applied to indicate significant difference. All statistical procedures were performed in SPSS 20.0.

4 Results

4.1 Soil moisture content and porosity

There are differences in soil moisture content under four different shrub restoration types. With the increase of depth, the soil moisture content of the same shrub rhizosphere do not change significantly (Fig. 3(a)), and the range of soil moisture content is 3.6–14.6%. Except for the 10 cm horizon, the moisture content of *L. bicolor* is significantly higher than other vegetation types. The moisture content of bare soil is less than that of shrub rhizosphere soil, but the difference is not significant. The soil moisture content decreased in the order *L. bicolor* > *Z. jujuba* > *P. sepium* > *P. granatum* > bare land. The soil porosity decreased significantly with increasing depth (Fig. 3(b)), and the porosity of deeper soil is significantly higher than that of surface soil. The variation of soil porosity ranged from 43 to 55.2%, and the order of average porosity of soil profiles of different land use types is: *P. sepium* > *Z. jujuba* > *P. granatum* > *L. bicolor* > bare land.

4.2 Moisture infiltration

Based on the dye tracer infiltration experiments on the four types of vegetation restoration and bare land, we obtain the black-and-white diagrams of the dyed profile and the change curve of the proportion of the dyed area with soil depth (Fig. 4). All results show a trend of decreasing staining area with increasing soil depth. In the *P. sepium* section (Fig. 4(a-1, a-2)), the dyed area represent 52.45% of the total area. The dyeing depth is 27.5 cm. The width of the dyeing area is max at 2.5–7.5 cm, and the dyeing ratio is 100%. In the *L. bicolor* section (Fig. 4(b-1, b-2)), the dyeing area percentage is 38.70% of the total area. The dyeing depth is 23.5 cm. The maximum staining width appeared at 3 cm, and the maximum staining ratio was 90.69%. In the *Z. jujuba* section (Fig. 4(c-1, c-2)), the dyeing area is 35.71% of the total area. The rate of decline in the proportion of dyed area increased significantly at 12.5 cm. The maximum dyeing width appeared at 10–13 cm, and the maximum dyeing ratio is 97.90%. No dyeing occurred at 27.5 cm. In the *P. granatum* section (Fig. 4(d-1, d-2)), the dyeing area is 49.92% of the total area. The dyeing depth is 30 cm. There is a maximum dyeing width at 6 cm, and the maximum staining ratio was 97.78%. Moisture tends to flow downwards. The bare land profile (Fig. 4(e-1, e-2)) had a dyeing ratio of 35.28%. The rate of decline in stained area increase sharply at 12.5 cm, and no stained plaque appeared at 22.5 cm. The maximum dyeing width is 7.5 cm, and the dyeing ratio is 100%. Compared with the experimental group with shrub roots, the tendency of water flow in the bare soil dyeing experiment is higher in the vertical.

4.3 Heterogeneity of spatial distribution of root system

The total RAR, the fine root, medium root and thick root area ratios of the corresponding sections of each vegetation are obtained by the profile method (Fig. 5). Except for soil horizon (10–20 cm), the RAR of *P. sepium* roots is significantly higher than that of other shrubs (Fig. 5(a)). In the vertical direction, the variation of RAR value of each root system is not consistent. For *P. sepium*, the maximum RAR value is 0.035% in the 40 cm soil horizon and a minimum RAR value is 0.007% in the 20 cm soil horizon. In *L. bicolor*, root RAR increase and then decrease with increasing depth, reaching a maximum of 0.013% at 30 cm. The RAR value of the *Z. jujuba* root system is significantly lower than other soil horizons at 30 cm. The minimum RAR value is 0.002%, and the maximum value is 0.014%. The RAR of *P. granatum* showed no obvious variation with depth, and the maximum RAR is 0.021% in the 30 cm soil horizons.

The RAR of fine root (root diameter less than 0.5 mm) is shown in Fig. 5(b). *Lespedeza bicolor* has few fine roots throughout the section and *P. granatum* has only few fine

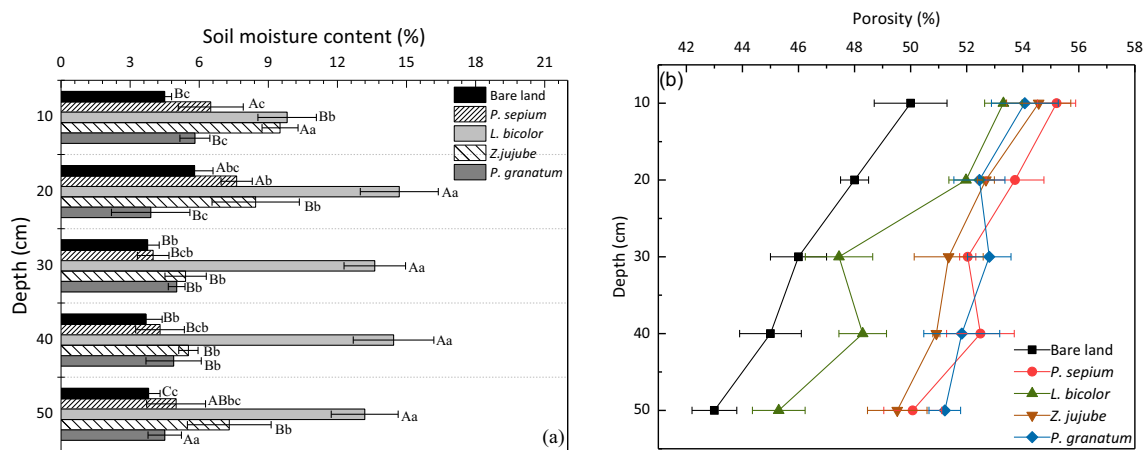


Fig. 3 Soil moisture content and porosity of bare land and four shrub rhizosphere soil. (a) is the evolution of soil moisture content with soil depth. (b) is the change of porosity with soil depth. Error bars denote standard errors of means ($n=3$). Different uppercase letters indicate

significant differences in the same land use type between different soil depths. Different lowercase letters indicate significant differences between different land use types. Samples with a common letter are not significantly different letters

roots in the 20 cm soil horizons. *Ziziphus jujuba* has more fine roots in uppermost 10 cm of soil, which decrease significantly with increasing soil depth ($p < 0.01$) and *P. sepium* has more fine roots in the 20–50 cm soil horizon.

The RAR of medium root (root diameter from 0.5 to 2 mm) is shown in Fig. 5(c). *P. sepium* and *Z. jujuba* have more medium roots in the 10 cm horizon. *Lespedeza bicolor* has more medium roots in the 20–30 cm soil horizon and *P. granatum* show a maximum at 30 cm. The medium roots RAR value of each vegetation type is significantly larger than the fine roots ($p < 0.01$).

The RAR of thick roots (root diameter greater than 2 mm) is shown in Fig. 5(d). The thick root RAR is the main component of the total RAR in the root system. The thick root RAR is significantly larger than that of fine roots and medium roots ($p < 0.01$). The thick root RAR of *P. sepium*, *Z. jujuba* and *P. granatum* all initially decrease, then increase and decrease again. There is a minimum value in the 20 cm horizon and a maximum RAR value in the 30–40 cm soil horizon. The *L. bicolor* thick roots RAR increase initially and then decrease with increasing depth, reaching a maximum at 30 cm.

4.4 Slope stability

Soil shear strength increases linearly with the increase of vertical positive pressure (Fig. 6), which follows the Mohr-Coulomb strength theory. Within the range of 10–30% of the set moisture content, the soil shear strength decreases with increasing moisture content. The maximum soil shear strength is reached at a moisture content of 10%. The shear strength decreases in the order *P. granatum* > *Z. jujuba* > *P. sepium* > *L. bicolor*.

The soil cohesion (C) and internal friction angle (φ) values of different shrubs at different soil moisture content are shown in Fig. 7. With the increase of soil water content, the C values

in the rhizosphere soil of the four shrubs decrease significantly, and the φ values decrease with the increase of soil moisture content, but there is no significant change. The maximum decrease in C is *P. sepium*, which decreases from 22.46 to 4.35 kPa, a decrease of 80.63%. *L. bicolor* has the largest decrease in friction angle in the rhizosphere soil. As the soil moisture content increases, φ decreases from 23.75° to 13.50°, a decrease of 43.16%. The C and φ values of *Z. jujuba* and *P. granatum* do not change significantly with the change of soil moisture content.

Using the C and φ values shown in Fig. 7, the safety factors for each slope are obtained (Fig. 8). Figure 9 is the x-direction displacement cloud diagram of the slope of the four shrub slopes when the safety factor is the maximum, that is, the slope at a soil moisture content of 10%. The change law of FoS is the same as the shear strength of the soil. And the greater the cohesion, the greater the FoS. The curved area is a potential sliding body. The smaller the potential sliding surface, the more stable the slope. The potential sliding area of *P. sepium* is the smallest value while slope FoS(1.39) is the largest at the same moisture content of 10%. When the soil moisture content is 30%, the FoS of *P. sepium* is reduced to 0.88, which is a decrease of 36.69%. With the increase of soil moisture content, the safety factor decreased, which was consistent with the change of C and φ . Under the condition of 20% moisture content, the safety coefficient of *P. granatum* has no change compared with the condition of 10% moisture. *Punica granatum* and *P. sepium* simultaneously achieve a maximum safety factor of 1.18 at this moisture content. Under the condition of 30% moisture content, the safety factors of the restoration slopes of the four shrubs were all less than 1. *Lespedeza bicolor* has the maximum value of 0.93. The safety factor of *L. bicolor* decreased by 25.6%, and it is least affected by soil moisture. At this moisture content, the safety coefficient of *P. granatum* decreases the most, down to

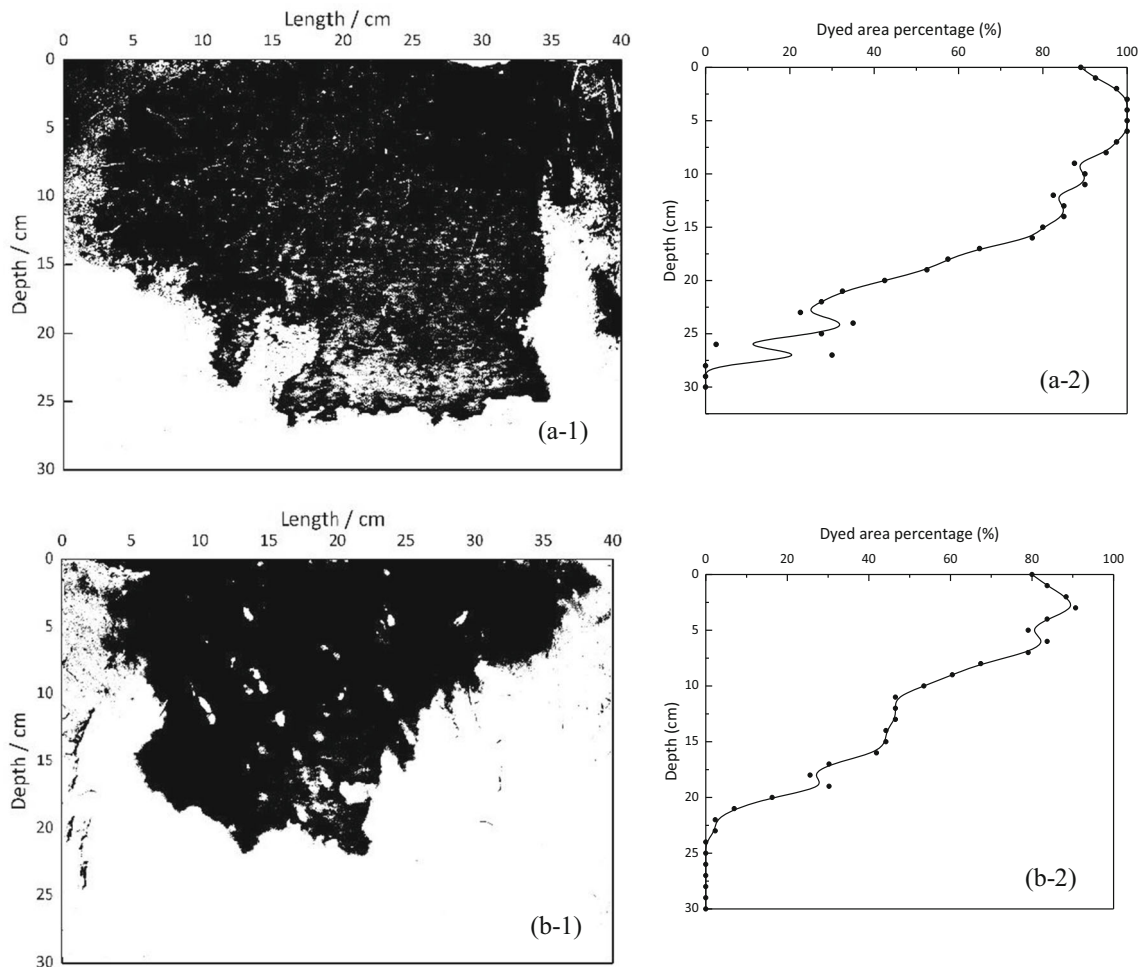


Fig. 4 Priority infiltration characteristics on bare land and four shrub profiles. 1 indicates the specific situation of the staining area on the set study section (30 cm × 40 cm). 2 represents the curve of the actual dyeing area ratio of the dye as a function of depth. (a-1) shows the distribution of the dyeing area of the *Punica. Sepium* (*P. Sepium*). (a-2) represents the relationship curve between actual dyeing area ratio and depth of the *Punica. Sepium* (*P. Sepium*). (b-1) shows the distribution of the dyeing area of the *Lespedeza. Bicolor* (*L. Bicolor*). (b-2) represents the relationship curve between actual dyeing area ratio and depth of the *Lespedeza.*

Bicolor (*L. Bicolor*). (c-1) shows the distribution of the dyeing area of the *Ziziphus. Jujube* (*Z. Jujube*). (c-2) represents the relationship curve between actual dyeing area ratio and depth of the *Ziziphus. Jujube* (*Z. Jujube*). (d-1) shows the distribution of the dyeing area of the *Periploca. Granatum* (*P. Granatum*). (d-2) represents the relationship curve between actual dyeing area ratio and depth of the *Periploca. Granatum* (*P. Granatum*). (e-1) shows the distribution of the dyeing area of the bare land. (e-2) represents the relationship curve between actual dyeing area ratio and depth of the bare land

0.7 (40.67%), and this slope is thus more likely to lose stability when the soil moisture changes.

5 Discussion

5.1 The relationship between plant roots and soil moisture

In this study, soil moisture content and porosity are larger than bare land, after the slopes had been planted with shrubs. The porosity of the deeper soil is significantly improved compared with the surface soil along the soil profile. *Lespedeza bicolor* has the greatest effect on improving soil moisture content, which is 2–4 times higher than that of bare land. *Periploca*

sepium has a strong effect on improving soil porosity, which increases by 5–10% compared with bare soil. Combined with the heterogeneity of the spatial distribution of plant roots, it is found that the RAR of *P. sepium* roots is significantly higher than that of other shrubs. The spatial distribution of plant roots has a certain effect on soil porosity. Plant roots can improve soil porosity. When the depth is relatively large, the RAR of the roots of the four types of shrubs reaches the maximum. It is believed that coarse roots will have a certain effect on soil porosity. The study of Gonzalez-Ollauri and Mickovski (2017) also shows that the plant root system improves the soil pore structure. This is related to the fact that plant root exudates can change the soil organic matter content, thereby improving soil structure and changing soil porosity (Gonzalez-Ollauri and Mickovski 2017). In addition, plant roots can

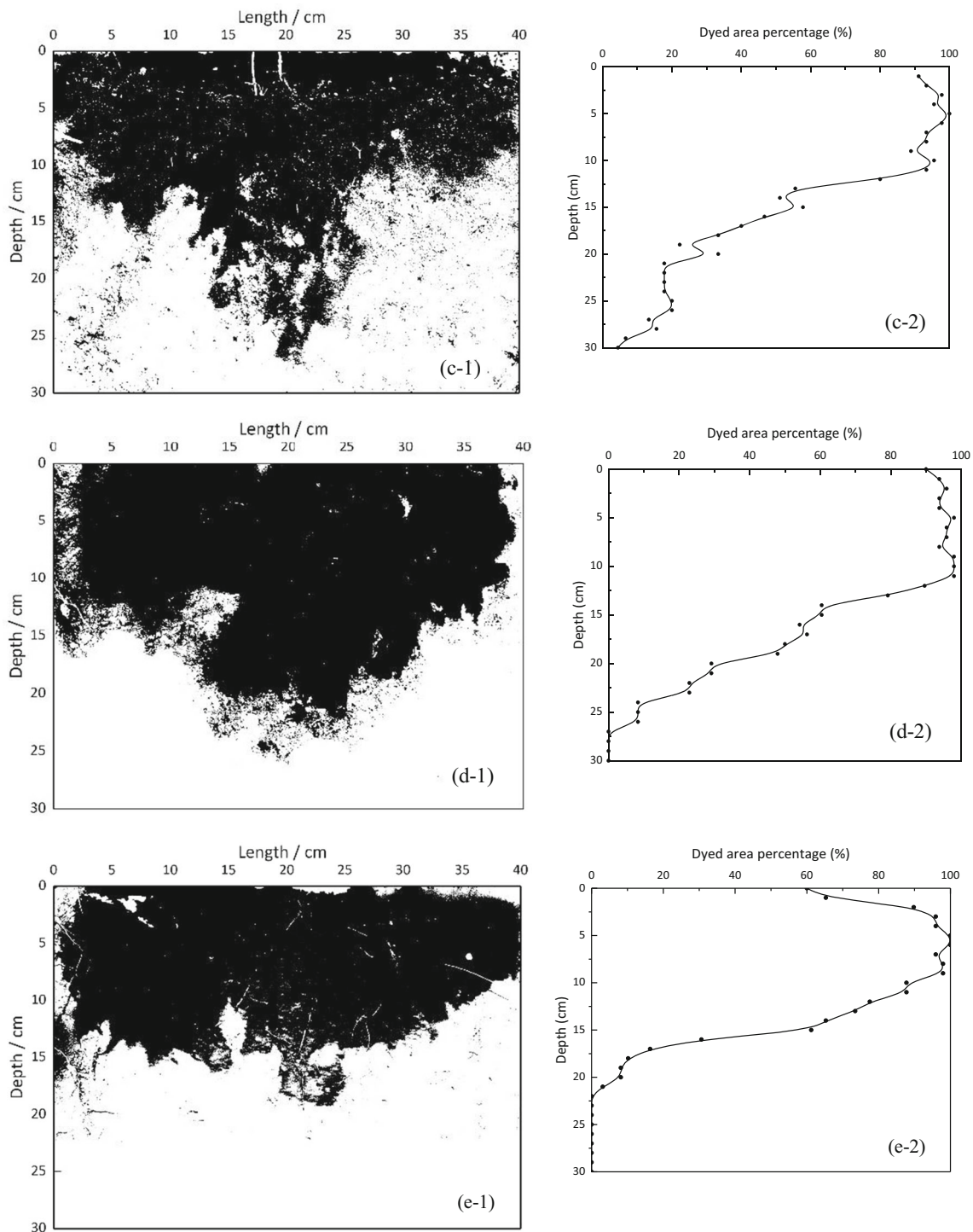


Fig. 4 (continued)

penetrate into shallow soils to form water channels, thus affecting soil moisture content and porosity (Boyle et al. 1989). Thick roots play an active role in forming water channels and increasing rhizosphere soil porosity, and are related to pores formed by soil particles. When the soil pore space is small, thick roots cannot grow in it, or the roots of the plant can't get

thick while they're growing (Wilson et al. 1977). Therefore, root diameter and soil moisture content and porosity affect each other. Moisture content and porosity affect root growth morphology and spatial location. The root system also changes the water channel and porosity of the soil, especially the thick roots.

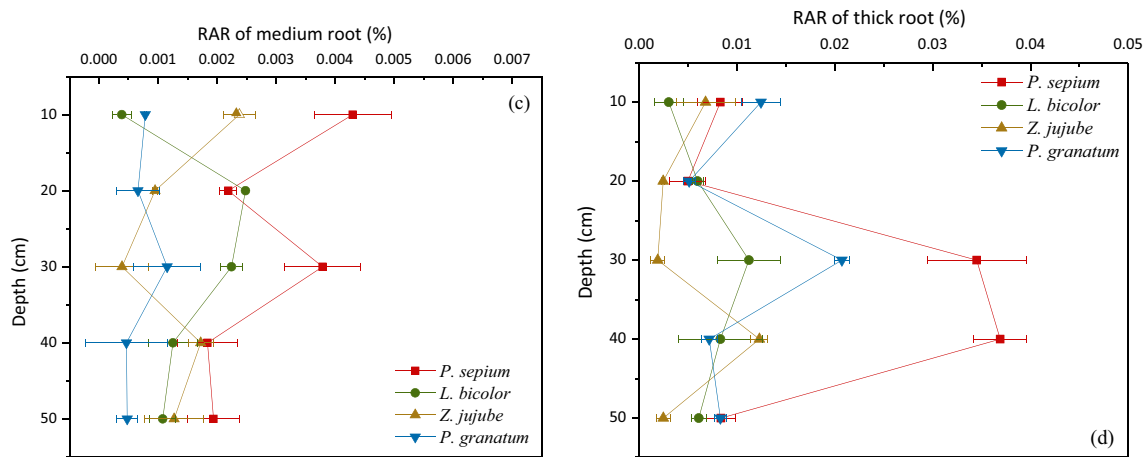


Fig. 5 (a) indicates the total RAR of four shrub plants. Error bars denote standard errors of means ($n=3$). Different uppercase letters indicate significant differences between each layer. Different lowercase letters indicate significant differences between different categories. Samples with a common letter are not significantly different letters. (b) indicates

the RAR of fine root (root diameter < 0.5 mm) of four shrub plants. (c) indicates the RAR of medium root ($0.5 \text{ mm} < \text{root diameter} < 2 \text{ mm}$) of four shrub plants. (d) indicates the RAR of thick root (root diameter $> 2 \text{ mm}$) of four shrub plants. Error bars denote standard errors of means ($n=3$)

The four vegetation restoration models and the dyed profiles of bare land all increased with depth. The width of the dyeing continued to decrease, and the dyeing area decreased. No dyeing occurred below 30 cm, and about 70% of the dyed area is concentrated above 15 cm in the soil horizon. The specific dyeing parameters are related to the vegetation

characteristics and soil properties. Jiang et al. (2017) study find that trees and ground cover systems are conducive to improve soil properties, as they can promote the formation of three-dimensional underground soil spaces, and have a positive effect on water redistribution. Nespoulous et al. (2018) study the subsurface preferential flow process and

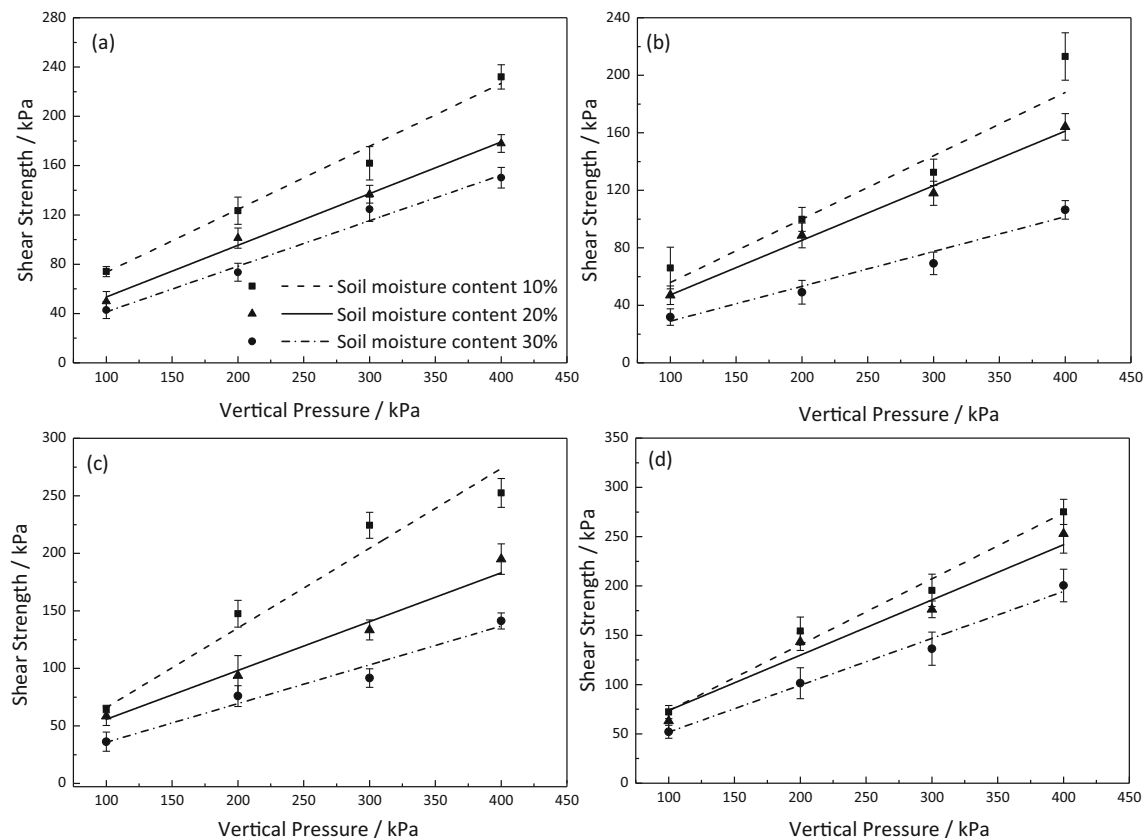


Fig. 6 The relationship between shear strength and vertical pressure of soil under three moisture content. Soil moisture content are 10%, 20%, and 30%. (a) *Punica. Sepium* (*P. Sepium*), (b) *Lespedeza. Bicolor* (*L. Bicolor*), (c) *Ziziphus. Jujube* (*Z. Jujube*), (d) *Periploca. Granatum* (*P. Granatum*)

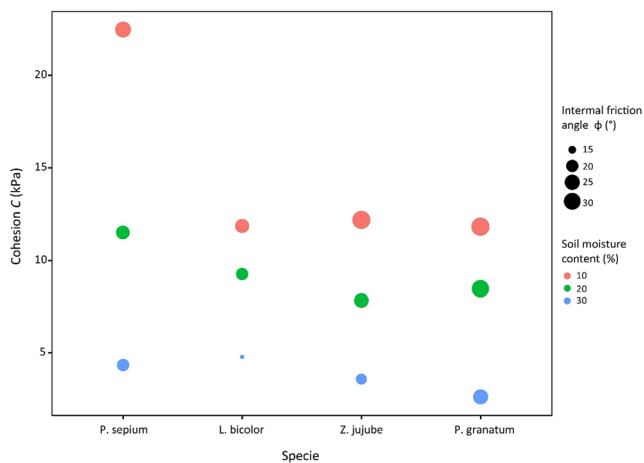


Fig. 7 Cohesion and internal friction angles of different plant at different soil moisture content. Points of different colors represent different soil moisture. The orange represents 10% moisture content, the green represents 20% moisture content, the blue represents 30% moisture content. The size of the point represents the magnitude of the internal friction angle, and the larger the diameter of the point, the larger the internal friction angle

individual plant soil profiles microscopically and macroscopically. The results show the complexity of the driving factors for optimal flow in the soil, in which fine roots promote the preferential flow.

The slopes of shrubs have better infiltration ability than bare slopes. Other previous research (Belsky et al. 1993; Van Schaik 2009; Perkins et al. 2012) also has shown that woodland soil has better permeability than grassland soil. The maximum width and the maximum dyeing area of each dyeing profile were mainly concentrated in the topsoil, which has high soil porosity. Combined with the spatial distribution of plant roots, fine roots (root diameters less than 0.5 mm) have high RAR values in topsoil (10–20 cm), which is conducive to the downward flow of water

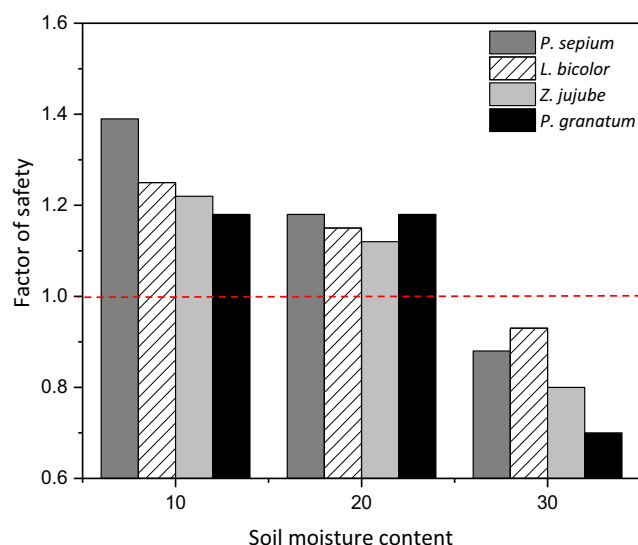


Fig. 8 Safety factor of slope under different water content

(Bodhinayake and Cheng Si 2004). The preferential flow area decreased gradually below 20 cm, and disappeared around 30 cm. Thick roots are mainly concentrated in this area. The RAR value of fine root and middle root is smaller than that of thick root. Therefore, we suggest that fine and middle roots form the main channels in the formation of the preferential flow, and good porosity is also favorable for water infiltration. Jiang et al. (2018) consider corn fiber root as the main channel of preferential flow. Nespoulous et al. (2018) also show that live fine roots have a greater effect on underground flow than thick roots and soil animals. Due to the fast growth of fine roots, a large number of non-capillary channels can be formed in a short period of time, thereby improving soil infiltration capacity. Underground runoff, especially preferential flow, can reduce the accumulation of interstitial water pressure and reduce the risk for landslides, which may be induced by rainfall (Nespoulous et al. 2018). Previous research has also suggested that the use of mixed shrubs will help control possible slope instability (Jiang et al. 2017; Liu et al. 2019).

5.2 Effect of root spatial distribution on slope stability

Of the four studied shrubs, *P. sepium* roots had the largest RAR. Thus, *P. sepium* can be considered a high-density root tree species, and can play a good role in stabilizing slopes. Preti and Giadrossich (2009) show that the more roots in a specific soil block, the more vegetation on the slope, and the better stability of the slope. Almost no root system is found below 50 cm. Most of the fine roots are present in the surface horizon of the soil, and decrease sharply with increasing depth, which is consistent with the results obtained by Wang et al. (2019) in abandoned croplands in the loess gully region of China. In this study, there is no obvious regular distribution of the medium roots in the whole section, and most of the thick roots are concentrated in the soil horizon around 30 cm. Nespoulous et al. (2018) find that in the Nan Lin Shan mountain of Xishuangbanna, Yunnan, China, compared with other soil types, the surface 30 cm of forest soil contains the most thick roots. Because of the plasticity of plant roots, the growth distribution in soil is anisotropic. Tateno et al. (2004) find that this phenomenon is closely and negatively related to soil moisture content, and the amount of fine roots located in the surface horizon is large. In the current study, the soil moisture content below 30 cm is slightly lower than that of the topsoil, and there are more thick roots in this area. This is related to in the presence of thick roots, nutrients, and water are consumed in the area around the root system (Bengough 2003). Therefore, Stokes considers that the slope stability is best when the plant root system is developed and undisturbed (Stokes et al. 2008).

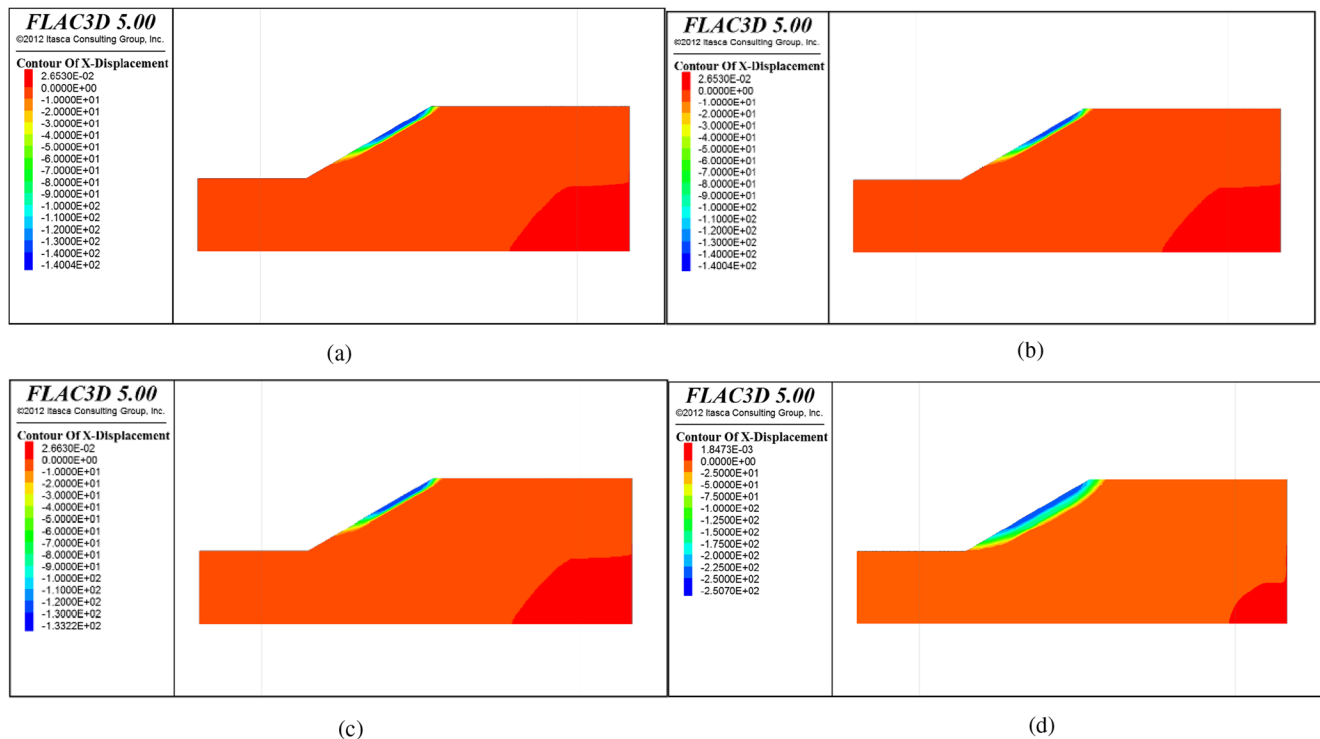


Fig. 9 Cloud diagram of slope displacement along the X direction when the four shrub slopes are at 10% moisture content. (a) *Punica. Sepium* (*P. Sepium*), (b) *Lespedeza. Bicolor* (*L. Bicolor*), (c) *Ziziphus. Jujube* (*Z. Jujube*), (d) *Periploca. Granatum* (*P. Granatum*)

5.3 Effect of soil moisture on slope stability

Soil moisture plays an important role in plant growth, root development, and soil strength (Whalley et al. 2008). In this study, three moisture conditions are set according to the natural moisture content and saturated moisture content, and the shear strengths obtained experimentally. After the moisture content increased, the shear strength decreases significantly, which is also found by Adekalu et al. (2007). Two important factors affecting soil shear strength are C and ϕ . Soil moisture content affects C and ϕ through the interaction between soil particles. The findings of the current study indicate that the rhizosphere C of *P. sepium* and the ϕ value of *L. bicolor* are significantly affected by soil moisture ($p < 0.01$). Increasing moisture content reduces the C value, so a decrease in C reduces the soil shear strength (Li et al. 2019a) and affects slope stability.

According to the C and ϕ values under different vegetation restoration modes, the corresponding FoS of the slope is obtained (Fattet et al. 2011). The FoS intuitively indicate the stability of the slope, which magnitude is the ratio of the sliding force to the sliding force along the specific assumed slip surface. When the safety factor is greater than 1, the slope is stable. When it is equal to 1, the slope is in the state of limit equilibrium. When it is less than 1, the slope is unstable. When the safety factor is large, the potential sliding surface is further away from the slope surface (Leshchinsky 1998). When the safety factor is greater than 1, the critical sliding surface is

considered to be the failure-sliding surface. When the safety factor is less than 1, the specific position of the sliding surface needs to be determined with a more accurate method (Li et al. 2019b). In this study, under the conditions of 10% and 20% soil moisture content, the safety factor of each slope is greater than 1, and at this time each slope is still in a stable state. Among these, the safety factor of the *P. sepium* slope is the largest at 10% moisture content, and thus planting *P. sepium* stabilized soil at 10% soil moisture content is considered optimal. In this study, at 10% soil moisture content, the FoS of the *P. sepium* is the largest, which shows *P. sepium* is considered an effective way to enhance slope stability. At a soil moisture content of 30%, the safety factors are all less than 1, and the slope is completely unstable at this time. The critical value of soil moisture content is between 20% and 30%. Combining soil moisture content, porosity, preferential flow, spatial distribution of root system, and slope safety factor, the findings of this study reveal that among the four shrub restoration types, *P. sepium* can achieve the best overall slope stabilization effect.

6 Conclusion

This study focuses on four different shrub species (*P. sepium*, *P. granatum*, *Z. jujuba*, and *L. bicolor*) for slope restoration in shallow landslide-prone areas. We used on-site inspection and soil dyeing experiments, the root area ratio is determined, and

shear strength tests of the remodeled soil are conducted. The interaction between soil moisture and plant roots and the effect of soil moisture on the shear strength as well as on slope stability of root soil are discussed. Our results suggest that the spatial distribution of soil moisture content is affected by factors such as plant species and root morphology. The soil moisture content and porosity of the shrub rhizosphere are improved compared with bare soil. *Periploca sepium* has the best effect on improving soil moisture content, and *P. granatum* has the best effect on improving soil porosity. The presence of plant roots (>2 mm) can significantly improve soil void structure and spatial distribution of soil moisture. The main channels for the formation of soil preferential flow are provided by fine and middle roots, and ameliorate soil porosity and infiltration. Sections with shrub root systems have preferential infiltration depths greater than those of bare land. The preferential flow area decreases continuously below 20 cm, and no dyeing occurred at 30 cm. According to the statistics of the root area ratio of roots, most of the fine roots were present in the soil surface horizon, which will decrease sharply with the increase of depth. There is no regular distribution of the medium roots throughout the entire section. Most of the thick roots were concentrated in the soil horizon around 30 cm. The plant roots and soil moisture parameters are closely related; soil moisture affect plant development and the spatial distribution of root systems. The increase of soil moisture content leads to a significantly decrease of the shear strength of the rhizosphere soil, which is not conducive to the stability of the slope soil. In summary, it is concluded that under the condition of natural soil moisture content and in areas where the soil moisture content does not change significantly, *P. sepium* has achieved the best slope protection and stability effect among the four shrub varieties studied. For areas with large changes in soil moisture content, *L. bicolor* shows good resistance to changes in soil moisture content and can play a better role in stabilizing slopes. In addition, plants with more middle and fine roots help to provide more water channels, promote soil infiltration, and improve slope stability. This result can provide theoretical support for the design of ecological restoration strategies in shallow landslide-prone areas.

Abbreviations FLAC3D, Fast Lagrangian Analysis of Continua 3D; φ , Internal friction angle; *L. bicolor*, *Lespedeza bicolor*; *P. sepium*, *Periploca sepium*; *P. granatum*, *Punica granatum*; RAR, Root area ratio; C, Soil cohesion; *Z. jujuba*, *Ziziphus jujuba*

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11368-021-02957-4>.

Funding Financial support for this research was provided by the National Natural Science Foundation of China Grant (NSFC 41971051) and the second Tibetan Plateau Scientific Expedition and Research (2019QZKK0603).

Data availability All data generated or analyzed during this study are included in this article.

Declarations

Conflict of interest The authors declare no competing interests.

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